Reminders

Use your AHEAD mail Id for Live Session otherwise the attendance will be marked as "Absent"

Submit all the Quizzes before the due date

[Both are the main eligibility criteria]

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Introduction to files

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Objectives

- Introduction to files and file organization.

Introduction to File

- File is a collection of records which are logically related to some object.
- Record value can be any form of data like numeric, character, date, string etc.
- For example: Student records are having values like Roll Number, Name, Date of Birth, Class.
- Files are usually stored in the secondary storage of a computer.

Types of File

Physical File

- Physical file contains the actual data that is stored.
- It also stores description about how the data is to be presented.

Logical File

- Logical file usually doesn't have actual data.
- It contains a description of records that are found in one or more physical files.
- It is a view or representation of one or more physical files.

File Types Based on Records

Fixed Length Record File

- Every record has same size (in bytes).
- Fixed sized memory blocks are assigned for each record as shown below:

- Advantage: Fast search of records.
- Disadvantage: Unnecessary wastage of memory

File Types Based on Records

Variable Length Record File

- Every record has variable size (in bytes).
- Variable sized memory blocks are assigned for each record as shown below:

- Advantage:
  - Efficient use of memory to store records.
  - Can move or save much faster than fixed sized

- Disadvantage: Slow access to records due to varying size of records.

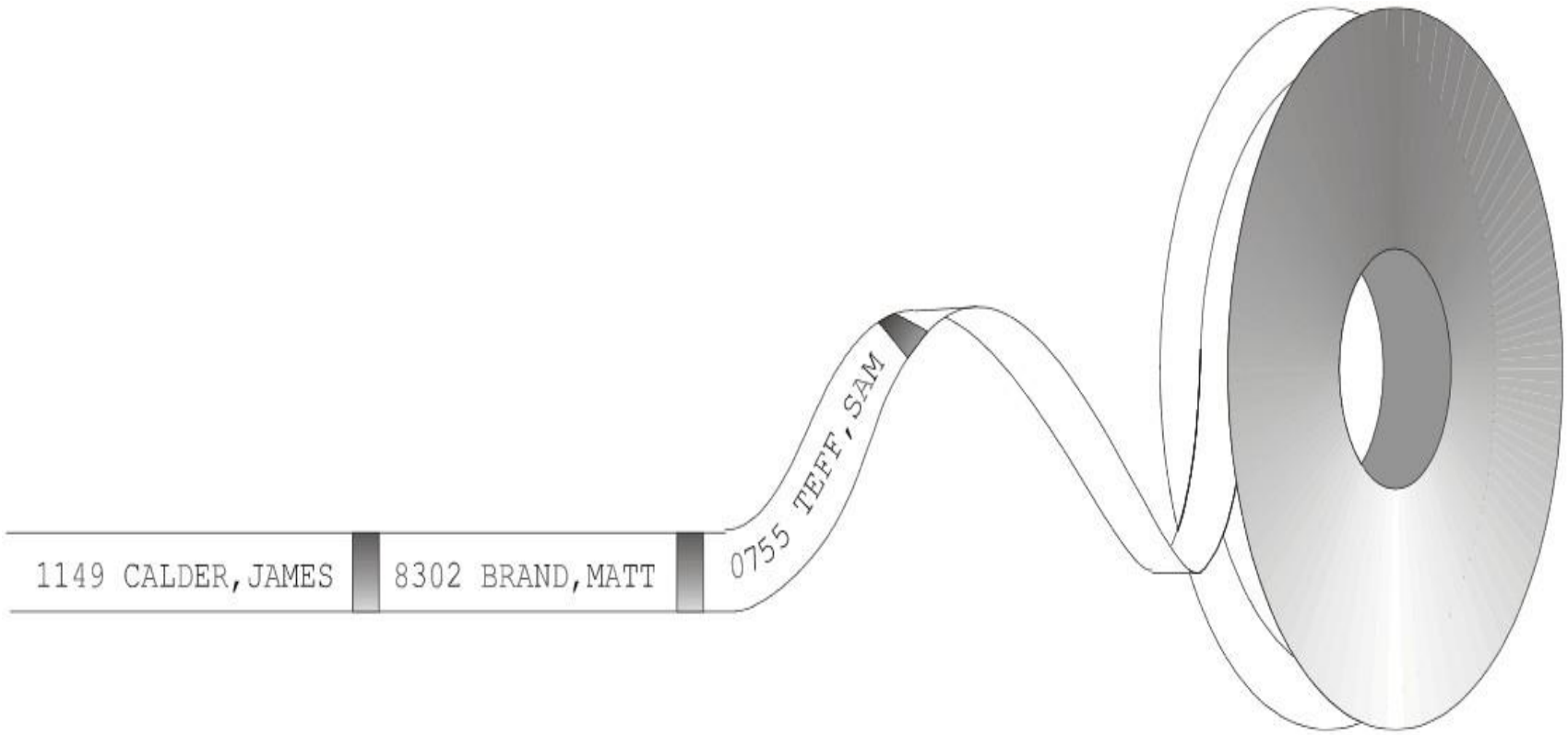
File Organization

- Storing the files in certain order is called file organization.
- Types of file organizations:
  1. Serial file organization
  2. Sequential file organization
  3. Indexed Sequential Access Method (ISAM)
  4. Direct access file organization

Serial File Organization

- Serial file organization is the simplest file organization method.
- In serial files, records are entered in the order of their creation.
- As such, the file is unordered, and is at best in chronological order.
- Serial files are primarily used as transaction files in which the transactions are recorded in the order that they occur.
- Serial files are also known as heap file.

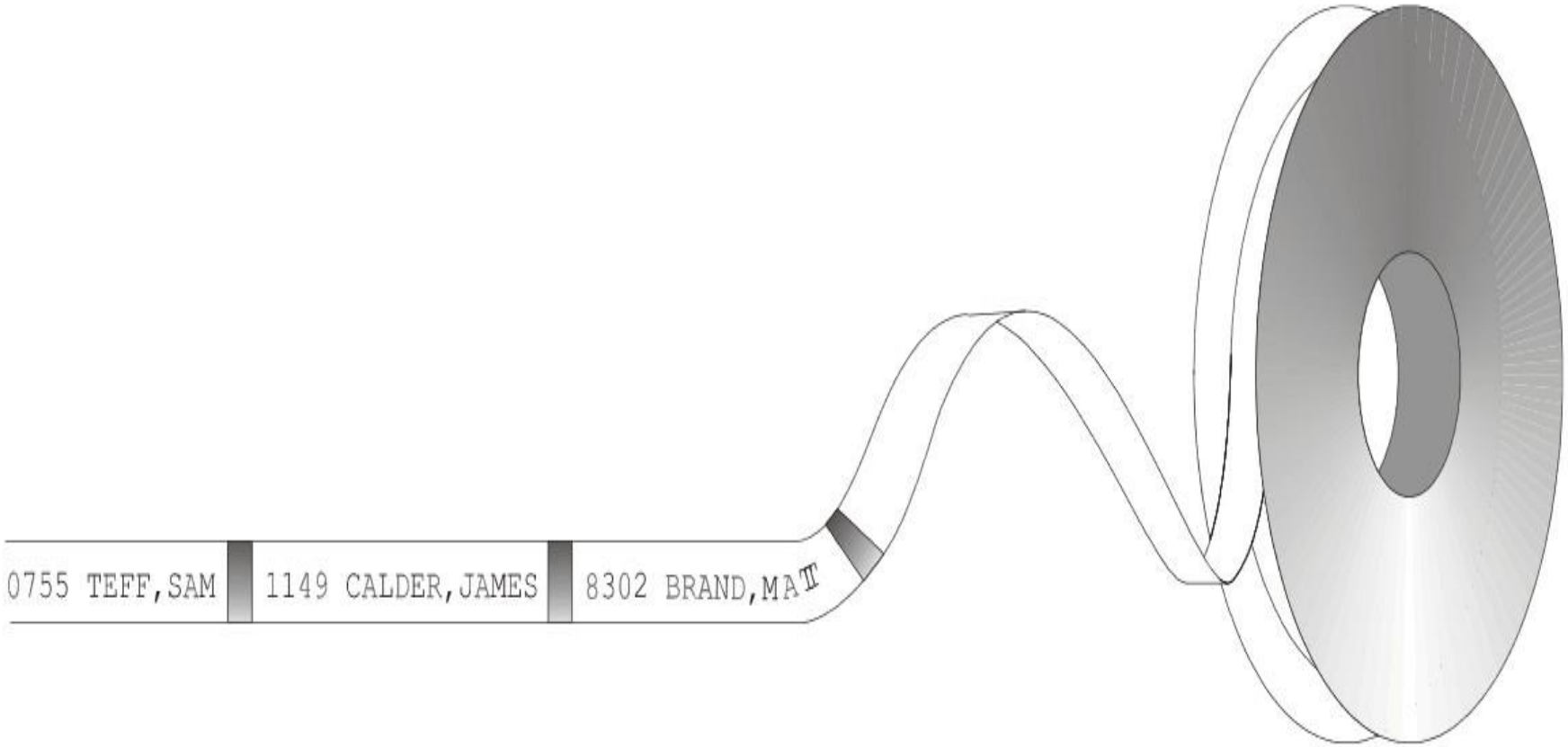
Serial File Organization



Sequential File Organization

- Sequential files are serial files whose records are sorted and stored in an ascending or descending on a particular key field.
- The search starts from the beginning of the file and the records can be added at the end of the file.
- Used in applications where there is a high file hit rate.
- Ideal for master files and batch processing applications.

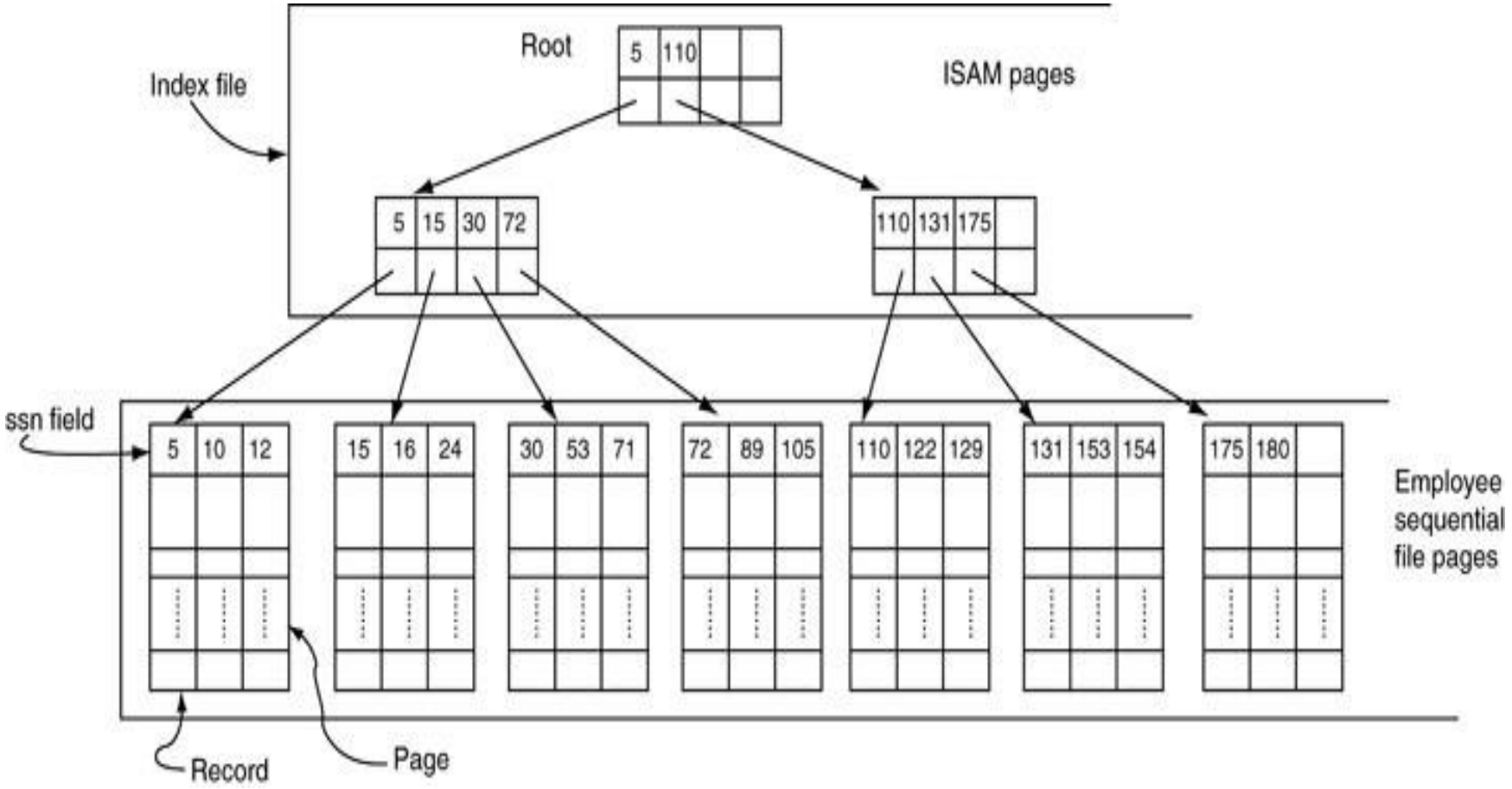
Sequential File Organization



Indexed Sequential File Organization

- Indexed sequential access file combines both sequential file and direct access file organization.
- Records are stored randomly on a direct access device such as magnetic disk by a primary key.
- An index value is mapped with every record.
- The data can be access either sequentially or randomly using the index.

Indexed Sequential File Organization



Direct access file organization

- Direct access file is also known as random access or relative file organization.
- The records are stored in direct access storage device, such as hard disk where it is randomly placed throughout the file.
- The records does not need to be in sequence because they are updated directly and rewritten back in the same location.
- Useful for immediate access to large amount of information. It is used in accessing large databases.
- It is also called as hashing.

Takeaways

- File system and the different types of files.
- File organizations.

Reference & Image Courtesy

- J. Glenn Brookshear, “Computer Science: An Overview”, Addision-Wesley, Twelfth Edition, 2014.
- 'A' Level Computing By P. M. Heathcote 2005.
- creativecommons.org

Objectives

- Need for structured data representation.

Introduction

- Data is fundamental to any computational task.
- The way by which data is collected, processed, and analyzed are depends on its format.
- A data's "structure" refers to a particular way of organizing and storing it in a database or warehouse so that it can be accessed and analyzed.
- Mainly two format:
  - A structured or
  - An unstructured format.

Unstructured vs Structured Data

Structured data

- It is comprised of clearly defined data types whose pattern makes them easily searchable.

Unstructured data

- "Everything else"
  – is comprised of data that is usually not as easily searchable, including formats like audio, video, and social media postings.

Structured Data

- In structured, the data is highly organized and neatly formatted.
- It is organized in tabular format (i.e. rows and columns) and there is a relationship between different rows and columns.
- As such, it's highly organized and formatted and easy to store, process, and access.
- Requires less storage space.
- Some examples of structured data are Excel files, Google Sheets, and traditional Database Management Systems (DBMS).

Structured Data

Pros :

- Universally understood
- Transferable to data tools
- Easily digestible for data programs
- Space savings

Cons

- Storage inflexibility
- Limited use cases

Unstructured Data

- The data which is not organized in any predefined manner.
- It can be textual, numbers, dates, or BLOBs (Binary Large OBjects).
- Irregularities and disorganization within unstructured data make it difficult to handle and understand.
- Some examples include text data, social media comments, documents, phone call transcriptions, various log files like server logs, sensor logs, image, audio, video, etc.
- The amount of unstructured data is much larger than that of structured data.

Unstructured Data

Pros :

- Wider use cases
- Flexible formatting
- Easy storage
- More data, more insights

Cons

- Difficult to prepare and analyze.
- It requires specific data tools.

Semi Structured Data

- It's a type of structured data that does not fit into the formal structure of a relational database.
- It is also referred to as data with a self-describing structure.
- A typical example of semi structured data is smartphone photos.
- Semi-structured data formats include JSON, CSV, and XML file types.

Takeaways

- Two categories of data
  - Structured Data
  - Unstructured Data
- Semi structured Data

Reference

1 J. Glenn Brookshear,"Computer Science: An Overview", Addision-Wesley, Twelth Edition, 2014

2 https://www.datamation.com/

3 RamezElmasri and ShamkantNavathe, "*Fundamentals of Database Systems*", Sixth Edition, Addison Wesley, 2010

4 Database Systems Concepts; Silberschatz, Abraham, Henry F. Korth, and S.Sudarshan

Objectives

- Introduction to Database Management System

Introduction

- Success of any organization depends on efficient use of its resources.
- Following are the major resources of any organization:
  - Buildings
  - Factories
  - Equipments
  - Human resources
  - Data

- How organizations process huge amount of data?
- How can they store it all?
- How they retrieve the data?

Databases

Data Vs Information

- Data constitute the building blocks of information.
- Information is produced by processing data.
- Information is used to reveal the meaning of data.
- Accurate, relevant, and timely information is the key to good decision making.
- Good decision making is the key to organizational survival in a global environment.

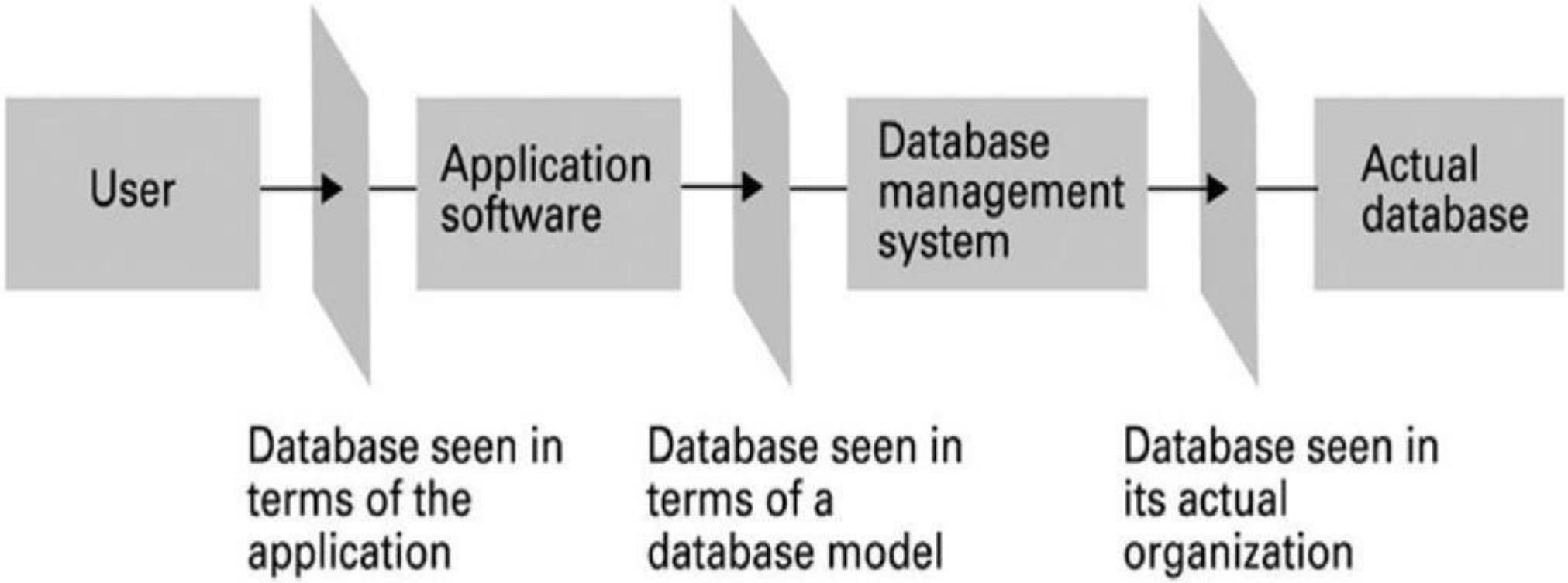
Introducing the Database

- A database is a shared, integrated computer structure that stores a collection of:
  - End-user data, that is, raw facts of interest to the end user.
  - Metadata, or data about data, through which the end-user data are integrated and managed.

DataBase Management System

- A set of programs to manage those data – Management system
  - Convenient
  - Efficient
- Management involves
  - Defining structures for storage of data
  - Providing mechanisms for the manipulation of data
  - Ensure the safety of the data stored
    - System crashes
    - Unauthorized access

System


User
Application software
Database management system
Actual database
Database seen in terms of the application
Database seen in terms of a database model
Database seen in its actual organization

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Database Management Systems - Examples

- MySQL – Free, Open Source RDBMS
- PostgreSQL – Free, Open Source Object relational system
- Microsoft Access – Part of MS Office suite
- MS SQL Server – RDBMS of Microsoft
- Oracle – Multi-model DBMS owned by Oracle Corporation

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Applications

- Enterprise Scenario
  - Sales: Customer, product, and purchase information
  - Accounting: Payments, receipts, account balances, assets
  - Human resources: Employee information, salaries, payroll taxes
  - Manufacturing: Supply chain Management, Production of items in factories, inventories of items in warehouses and stores, and orders for items
  - Online Sales: Online orders, recommendation lists, and online product evaluations

Database-System Applications (Contd)

- Banking and Finance
  - For customer information, accounts, loans, and banking transactions.
  - Credit card transactions: purchase details on credit cards and monthly statements
  - Finance: Sales, and purchase details of financial instruments such as stocks and bonds.

Database-System Applications (Contd)

- Health Care
  - Hospitals – Patients, Doctors, Staff, Appointments, Test Results
  - Pharmacy
  - Blood Bank

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Database-System Applications (Contd)

- Universities
  - For student information, course registrations, Exams and grades
  - Faculty information
  - Counselling
  - Achievement tracking
  - Leave Management
  - Accounting
  - Library
  - Hostel

Database-System Applications (Contd)

- Telecommunication
  - Records of calls made
  - Monthly bills
  - Balances on prepaid calling cards
  - Communication network details

Types of Databases

Databases can be classified according to;

- Number of Users:
  - Single-user database
    - Desktop database
  - Multiuser database:
    - Workgroup database
    - Enterprise database
- Data Location:
  - Centralized database
  - Distributed database
- Data Usage:
  - Operational database/ transactional database / production database
  - Data warehouse

Takeaways

- Data vs Information.
- DBMS introduction.

References

- Introduction to Database Systems Introduction to Database Systems (slidetodoc.com)
- Database Systems Concepts; Silberschatz, Abraham, Henry F. Korth, and S.Sudarshan
- Principles of Distributed Database Systems; Ozsu, M. Tamer and Patrick Valduriez.
- Database Systems - Design, Implementation, and Management (9th Edition) by Rob and Coronel.
- C. S. R. Prabhu, "Object Oriented Database Systems : Approache and Architectures", Third Edition, PHI Learning Pvt. Ltd.
- RamezElmasriand ShamkantNavathe, "*Fundamentals of Database Systems*", Sixth Edition, Addison Wesley, 2010

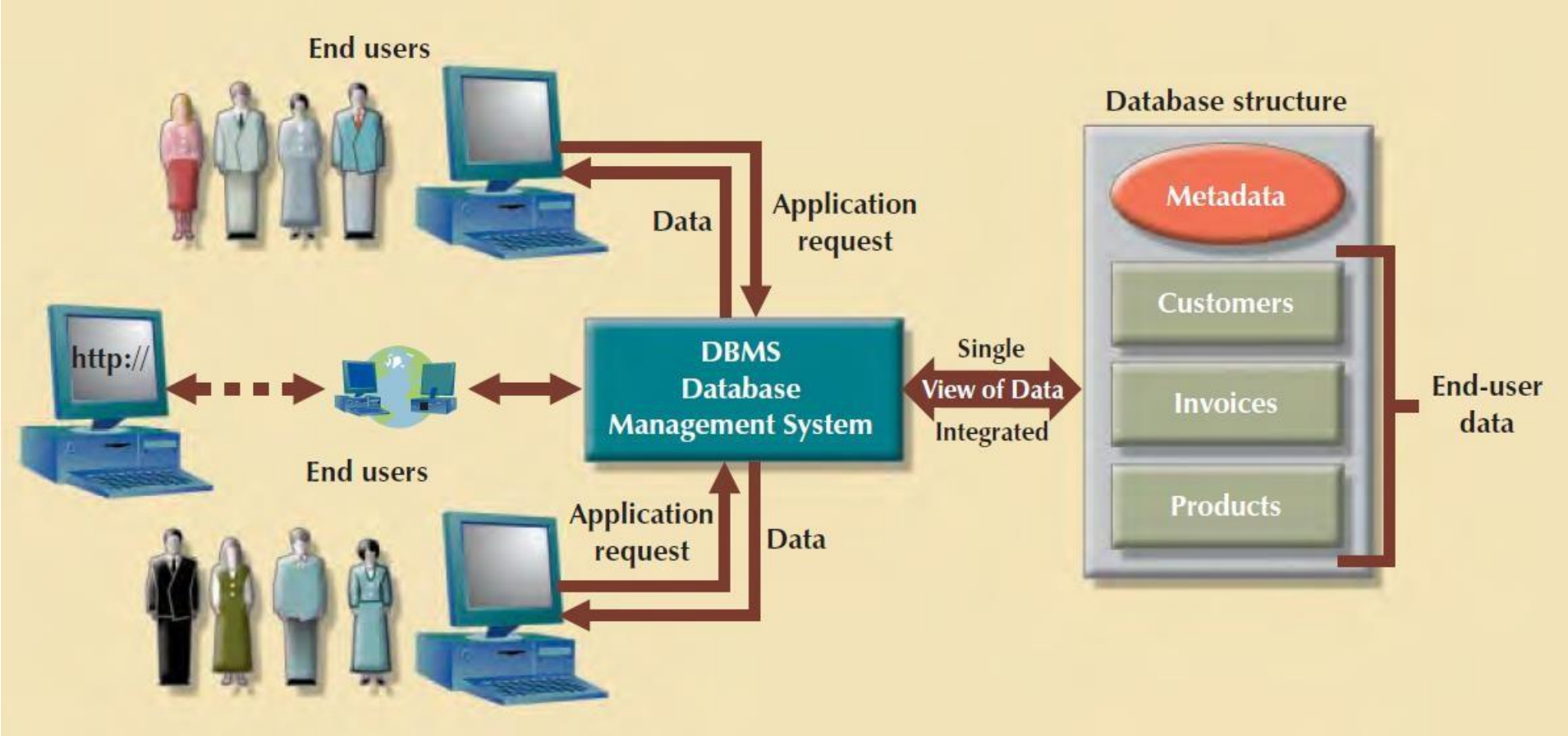
Objectives

- To get an overview on Database Management Systems (DBMS).
- Different Data models and why data models are important.

Database Management System (DBMS)

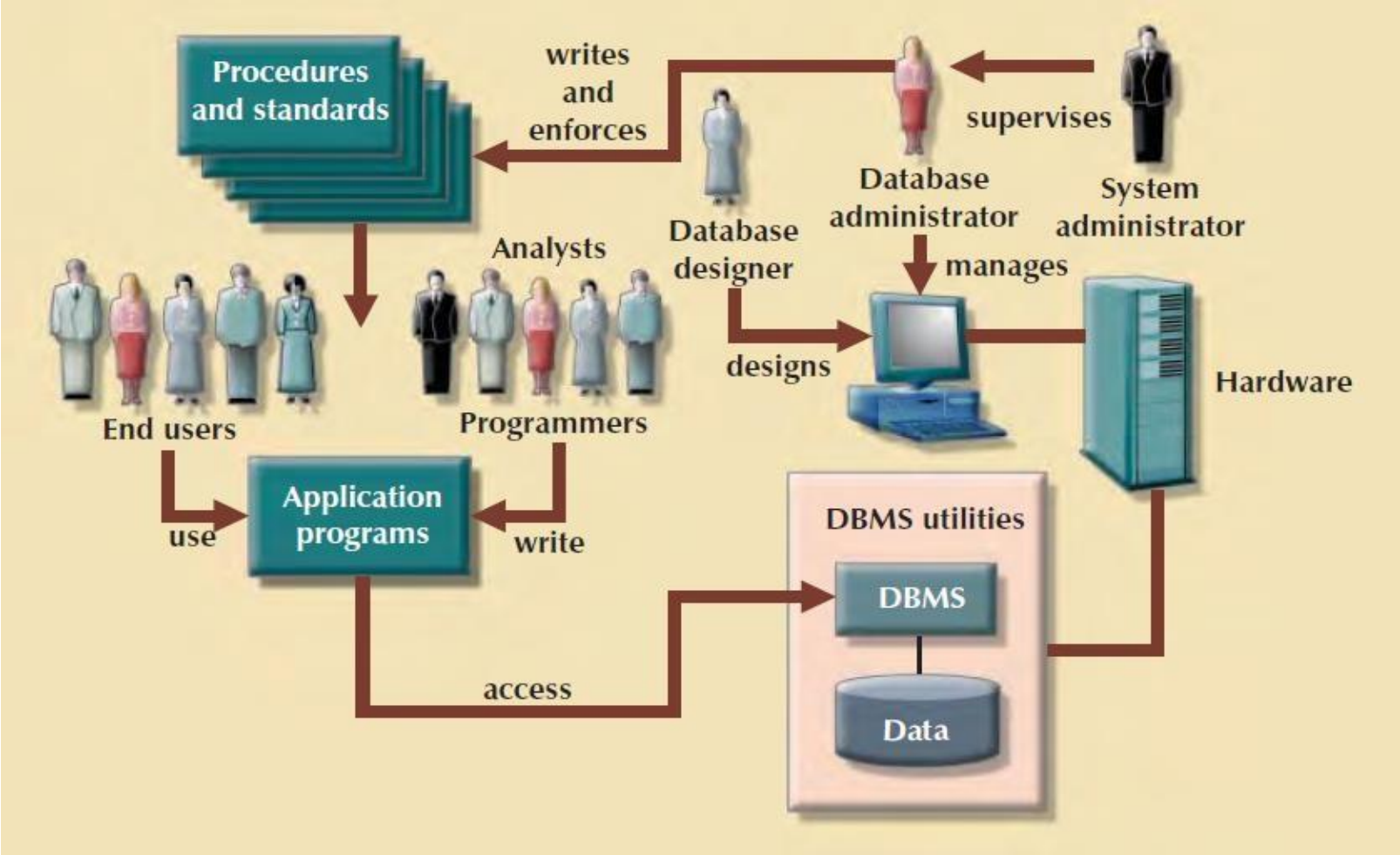
- A database management system is a collection of programs that manages the database structure and controls access to the data stored in the database.
- It serves as the intermediary between the user and the database.
- The database structure itself is stored as a collection of files, and the only way to access the data in those files is through the DBMS.

Role of DBMS



DBMS manages the interaction between the end user and the database

The Database System Environment



DBMS Functions

- A DBMS performs several important functions that guarantee the integrity and consistency of the data in the database.
- They include:
  - data dictionary management
  - data storage management
  - data transformation and presentation
  - security management
  - multiuser access control
  - backup and recovery management
  - data integrity management
  - database access languages and application programming interfaces
  - database communication interfaces

Advantages of DBMS

- Improved data sharing
- Improved data security
- Better data integration
- Minimized data inconsistency
- Improved data access
- Improved decision making
- Increased end-user productivity

Database Models

- Database model is the conceptual view of a database
- Evolution of Data Models:
  - Hierarchical
  - Network
  - Relational
  - Entity relationship
  - Object oriented (OO)
- Commonly used database models are
  - Relational database model
  - Object-oriented database model

Relational Model

- Introduced in 1970 by E. F. Codd (of IBM).
- The relational model foundation is a mathematical concept known as a relation.
- Each row in a relation is called a tuple.
- Each column represents an attribute.
- Also, describes a precise set of data manipulation constructs based on advanced mathematical concepts.

A Relation Containing Employee Information

Empl Id	Name	Address	SSN
25X15	Joe E. Baker	33 Nowhere St.	111223333
34Y70	Cheryl H. Clark	563 Downtown Ave.	999009999
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555
•	•	•	•
•	•	•	•
•	•	•	•

Relational Model

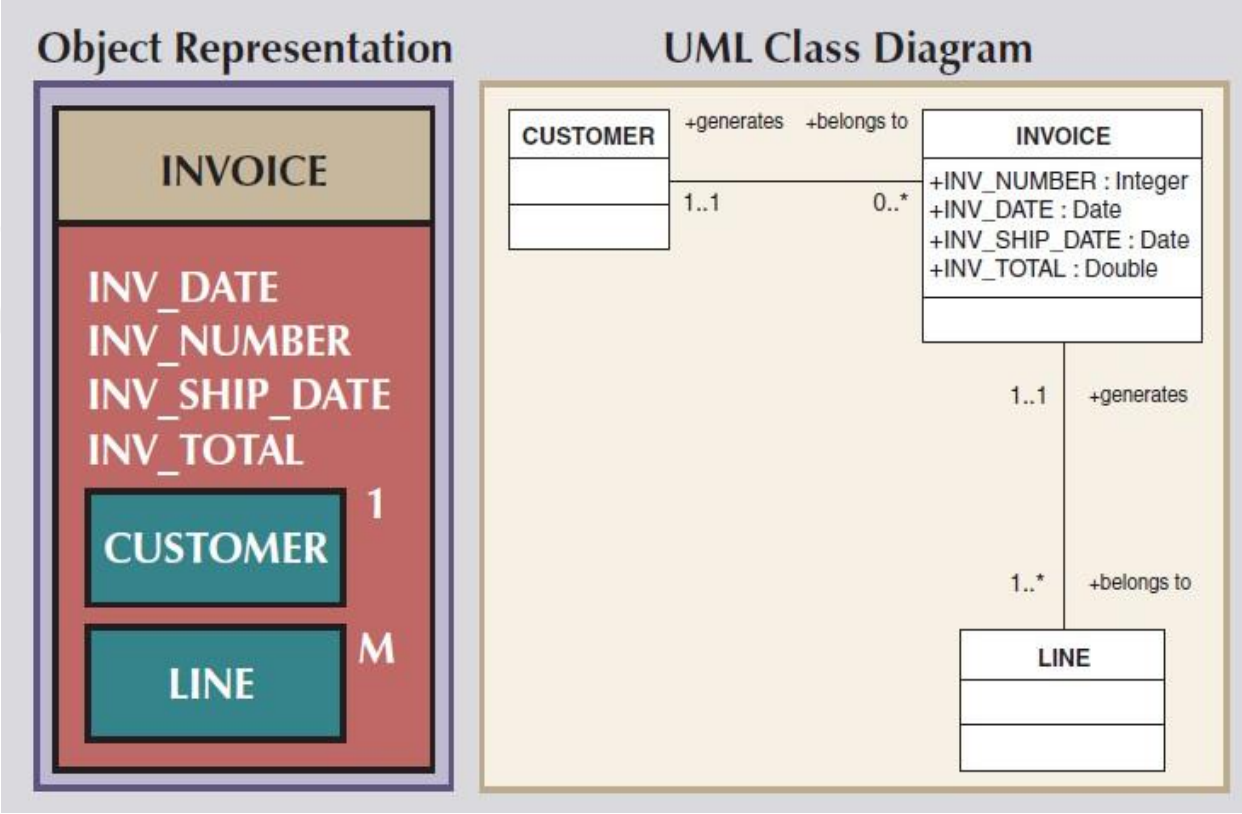
- Relational model is the basis of Relational Database Management System (RDBMS).
- RDBMS performs the same basic functions provided by the hierarchical and network DBMS systems
- Advantage of the RDBMS is its ability to hide the complexities of the relational model from the user.
- The relational data model's rise to dominance is its powerful and flexible query language : Structured Query Language (SQL).
- SQL retrieves data with far less effort than any other database or file environment.

Object Oriented Model (OODM)

- OODM is the basis for the object-oriented database management system (OODBMS).
- An OODM reflects a very different way to define and use entities.
- The OO data model is based on the following components:
  - An object is an abstraction of a real-world entity.
  - Attributes describe the properties of an object.
  - A class is a collection of similar objects with shared structure and behavior.
  - Classes are organized in a *class hierarchy*.
  - Inheritance is the ability of an object within the class hierarchy to inherit the attributes and methods of the classes above it.

Object Oriented Model

- Object-oriented data models are typically depicted using Unified Modeling Language (UML) class diagrams.
- **UML** is a language based on OO concepts that describes a set of diagrams and symbols that can be used to graphically model a system.



Summary

- Database Management System and its role.
- Functions & Advantages of DBMS.
- Different Database Models
- Relational DBMS
- Object Oriented DBMS

Reference & Image Courtesy

- J. Glenn Brookshear, "Computer Science: An Overview", Addision-Wesley, Twelfth Edition, 2014.
- Database Systems - Design, Implementation, and Management (9th Edition) by Rob and Coronel.